

EcoMule: Autonomous Rainforest Monitoring System with BLE-Triggered Data Mule and Cellular Telemetry

Pranav Vishal Deshmukh
B.Tech 3rd Year

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati
Guwahati, India
Email: pranav.deshmukh@iitg.ac.in

Dhruv Patel
B.Tech 3rd Year

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati
Guwahati, India
Email: u.patel@iitg.ac.in

Abstract—This paper presents a highly autonomous, solar-harvesting acoustic edge-computing node designed for prolonged deployment in deeply remote eco-environments. The architecture leverages an ESP32-S3 SoC interfaced with an INMP441 I2S MEMS microphone, executing real-time Digital Signal Processing (DSP) with Automatic Gain Control (AGC) to capture and store high-fidelity environmental audio on local flash memory. To circumvent the severe energy penalties associated with continuous wireless transmission, a Bluetooth Low Energy (BLE) “Data Mule” framework is introduced. This framework permits field researchers to asynchronously awaken an embedded Wi-Fi access point, facilitating bulk data extraction via a memory-safe, client-side .tar archiving protocol. Furthermore, the system integrates a 4G LTE-Cat-1 modem alongside an analog telemetry circuit to autonomously broadcast storage capacity and battery health via SMS. Packaged within a weather-resistant enclosure, this system delivers a robust, deploy-once solution for biodiversity tracking and anthropogenic threat detection in infrastructure-deficient topographies.

Index Terms—Eco-Acoustics, Data Mule, Digital Signal Processing (DSP), I2S, Edge Computing, Telemetry, ESP32-S3, BLE

I. INTRODUCTION

The continuous acoustic monitoring of remote rainforest ecosystems provides invaluable datasets for biodiversity assessment, avian migratory tracking, and the real-time detection of illegal logging activities. However, the deployment of electronic sensor nodes in such topographies presents severe engineering constraints: the absence of grid power, erratic cellular connectivity, and extreme environmental humidity [1].

Conventional Internet of Things (IoT) acoustic monitors attempt to stream heavy WAV files directly to cloud servers via cellular or Wi-Fi networks. This methodology induces rapid battery depletion, often rendering the devices defunct within days, particularly under dense forest canopies where photovoltaic (PV) yield is suboptimal.

To surmount these constraints, this project proposes a completely decentralized, edge-storage paradigm. Utilizing a 7Semi ESP32-S3 embedded framework [1], the device operates primarily in a low-power acoustic logging state. Data retrieval

is exclusively localized; utilizing a “Data Mule” topology, the system broadcasts a microwatt Bluetooth Low Energy (BLE) beacon. When a field ranger approaches, a localized BLE handshake triggers a temporary, high-throughput Wi-Fi interface for localized data offloading. To ensure administrative oversight without compromising power, an integrated Quectel EC200U LTE modem dispatches concise SMS telemetry bursts regarding energy and storage metrics at predefined temporal intervals [2].

II. HARDWARE ARCHITECTURE AND COMPONENTS

To guarantee signal integrity and mitigate electromagnetic interference (EMI) originating from the RF transceivers, the hardware topology enforces strict power domain segregation, as illustrated in Fig. 1.

As shown in Fig. 2, the physical layout consists of the following key subsystems:

- **Microprocessor: 7Semi ESP32-S3 EC200U LTE Cat-1.** Combines a dual-core ESP32-S3 with a Quectel EC200U 4G LTE module. It handles file system management, web hosting, and Machine-to-Machine (M2M) AT command execution.
- **Microphone: INMP441 Omnidirectional I2S MEMS.** A high-SNR digital microphone. By converting analog sound to a digital I2S stream directly at the capsule, it prevents long connecting wires from acting as antennas that absorb EMI from the Wi-Fi and Cellular radios [3].
- **Real-Time Clock: DS1302 Module.** Provides persistent timekeeping for precise file timestamping (YYYYMMDD_HHMMSS), backed by a CR2032 coin cell, ensuring temporal accuracy even during system reboots [4].
- **Energy Subsystem: Waveshare Solar Power Manager (D) & 10W 12V Solar Panel.** Utilizes Maximum Power Point Tracking (MPPT) to efficiently charge a 1S3P Samsung INR18650-35E Lithium-Ion battery pack (10 500 mA h capacity).

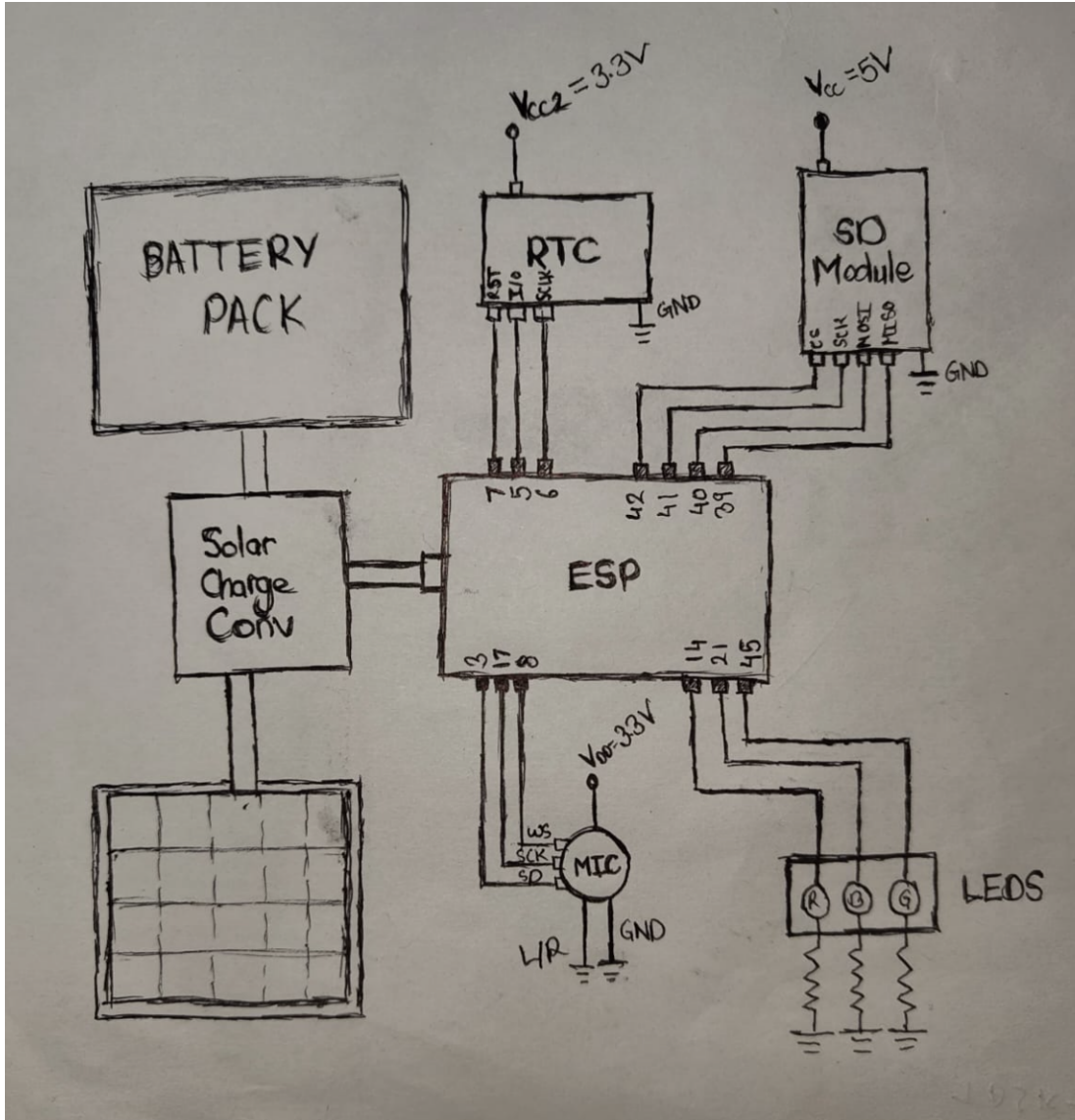


Fig. 1. Complete system architecture diagram illustrating the strict power domain segregation between the Waveshare MPPT solar manager, the ESP32-S3, and the highly sensitive I2S/SPI peripherals.

- **Visual Telemetry: 3-LED Debugging Interface.** Headless state indication via Green (Recording/Heartbeat), Blue (Comms/BLE/Wi-Fi), and Red (Hardware Faults) LEDs.

A. Headless Visual Telemetry Interface

To eliminate the reliance on serial monitors during field deployment, a three-tier LED diagnostic interface was integrated. A Green LED serves as a heartbeat indicator, flashing during active I2S-to-SD write cycles and confirming configuration updates. A Blue LED denotes wireless communication statuses, indicating BLE handshakes, active Wi-Fi states, and flashing rapidly during cellular SMS transmissions. Finally, a Red LED acts as a strict hardware fault indicator, illuminating only during critical failures such as an SD card physical disconnect or an I2S clock synchronization error.

III. SOFTWARE AND DSP IMPLEMENTATION

A. Direct Memory Access (DMA) & I2S Buffering

High-fidelity acoustic recording to SD cards frequently suffers from “stuttering” due to the variable latency of SPI flash block-erase cycles. To decouple the acoustic sampling from the SD write delays, the firmware allocates 8 hardware-managed DMA buffers (64 bytes each). This permits the I2S peripheral to continuously aggregate acoustic data in the background independently of the main CPU execution thread.

B. Digital Signal Processing & Smart AGC

The INMP441 transmits 32-bit digital packets, whereas the standard uncompressed WAV format utilizes 16-bit PCM encoding for optimal storage efficiency. Because rainforest acoustic environments are highly volatile—ranging from a faint

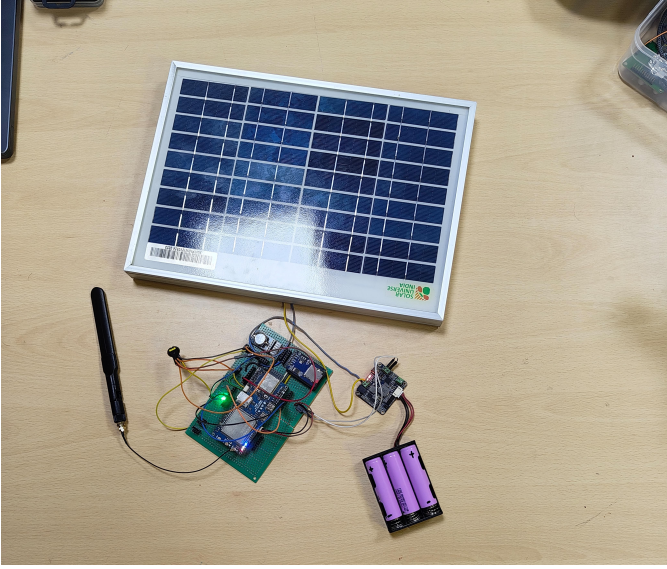


Fig. 2. The physical hardware prototype featuring the 7Semi ESP32-S3, INMP441 microphone, and DS1302 RTC wired on the test bench with the 3-LED visual telemetry interface.

bird call at 30dB to a deafening chainsaw at 110dB—a static DSP gain would cause severe distortion.

An Automatic Gain Control (AGC) DSP pipeline was engineered:

$$y[n] = \max \left(-32768, \min \left(32767, (x[n] \gg 14) \cdot G_{dyn} \right) \right) \quad (1)$$

The ESP32 analyzes the absolute peak amplitude of every incoming audio chunk. If the ambient acoustic floor drops, the microcontroller autonomously scales the floating-point multiplier (G_{dyn}) to “zoom in” on faint bio-acoustics. If a sudden transient noise occurs, the algorithm rapidly compresses the gain to prevent 16-bit integer overflow, ensuring captured datasets remain pristine for future AI classification.

C. Self-Healing Operating System

Early prototypes exhibited crashes due to Virtual File System (VFS) descriptor leaks and SD card SPI dropouts. The firmware was fortified with two major self-healing mechanisms:

- 1) **VFS Memory Management:** Explicit `file.close()` enforcement during asynchronous web-server loops guarantees that the ESP32 never hits its 5-file concurrent limit.
- 2) **Hardware Watchdogs:** If the I2S microphone clock disconnects, a 1-second timeout aborts the read to prevent an infinite freeze. If the SD card fails to mount, a non-blocking `SD.end()` and `SD.begin()` background loop continuously attempts hardware recovery.

D. Concurrent Execution and Interleaving

A significant software engineering hurdle was maintaining the Web Server’s responsiveness while the single-threaded recording loop processed massive I2S audio chunks. Implementing a strict Mutex (Mutual Exclusion) would have forced audio logging to pause during Ranger downloads. Instead, a

concurrent execution model was developed. By injecting the `HTTP server.handleClient()` routine directly into the localized audio-chunking `while`-loop, the ESP32 successfully interleaves HTTP packet processing with I2S DMA retrieval. This ensures seamless bulk TAR downloads without corrupting the simultaneous environmental acoustic recordings.

IV. DATA MULE & RETRIEVAL WORKFLOW

As conceptualized in Fig. 4, the system relies on physical transit to bridge deep-forest data gaps.

A. Asynchronous BLE State Machine

By default, the 802.11 Wi-Fi radio is completely powered down. The device broadcasts a 1 mW BLE GATT signature (UUID: 4fafc2...). Upon receiving a UTF-8 “WAKE” payload from a recognized ranger client, the ESP32 initiates a state transition, activating a captive Web Server. If no HTTP traffic is detected within a 300,000-millisecond threshold, the system autonomously deprecates back to the BLE-only sleep state.

B. Asynchronous UI and Client-Side Archiving

The Web UI (Fig. 5) is dynamically generated and streamed to the user in chunks using `server.sendContent()` to bypass the ESP32’s limited RAM. The dashboard provides full control to change recording durations, pause intervals, and SMS periods on the fly.

To facilitate the rapid extraction of dozens of audio files, a Javascript “Bulk Download” function was developed. Traditional browsers block multiple simultaneous file downloads for security reasons. To bypass this, the Javascript fetches the selected WAV files asynchronously into the smartphone’s RAM, compiles them into a single `.tar` POSIX archive mathematically on the client side, and triggers a single save prompt. Downloaded files are tracked using the browser’s `localStorage`, automatically applying a “Downloaded” badge to visually assist the ranger.

V. SYSTEM PERFORMANCE

As seen in Fig. ??, the cellular telemetry ensures that the device’s operational status is continuously verifiable without requiring a physical trek into the forest. Power profiling confirms the system’s viability for permanent deployment. Utilizing a 10 W monocrystalline panel and a 1S3P 10 500 mA h Samsung 35E energy reserve (38.8 Wh), the harvested energy easily exceeds the daily demand.

With a 116.5 Wh reserve capacity, the node achieves an autonomous persistence of > 14 days in the event of total solar obstruction (e.g., severe monsoon conditions).

A. Acoustic Fidelity and Storage Optimization

While the INMP441 natively outputs a 24-bit audio signal packaged within a 32-bit I2S frame, the system explicitly down-samples this to a 16-bit WAV file. A 16-bit depth provides 96 dB of dynamic range. According to the sensor’s datasheet, the INMP441 possesses an acoustic overload point of 120 dB SPL and a noise floor of 33 dB SPL, yielding an effective

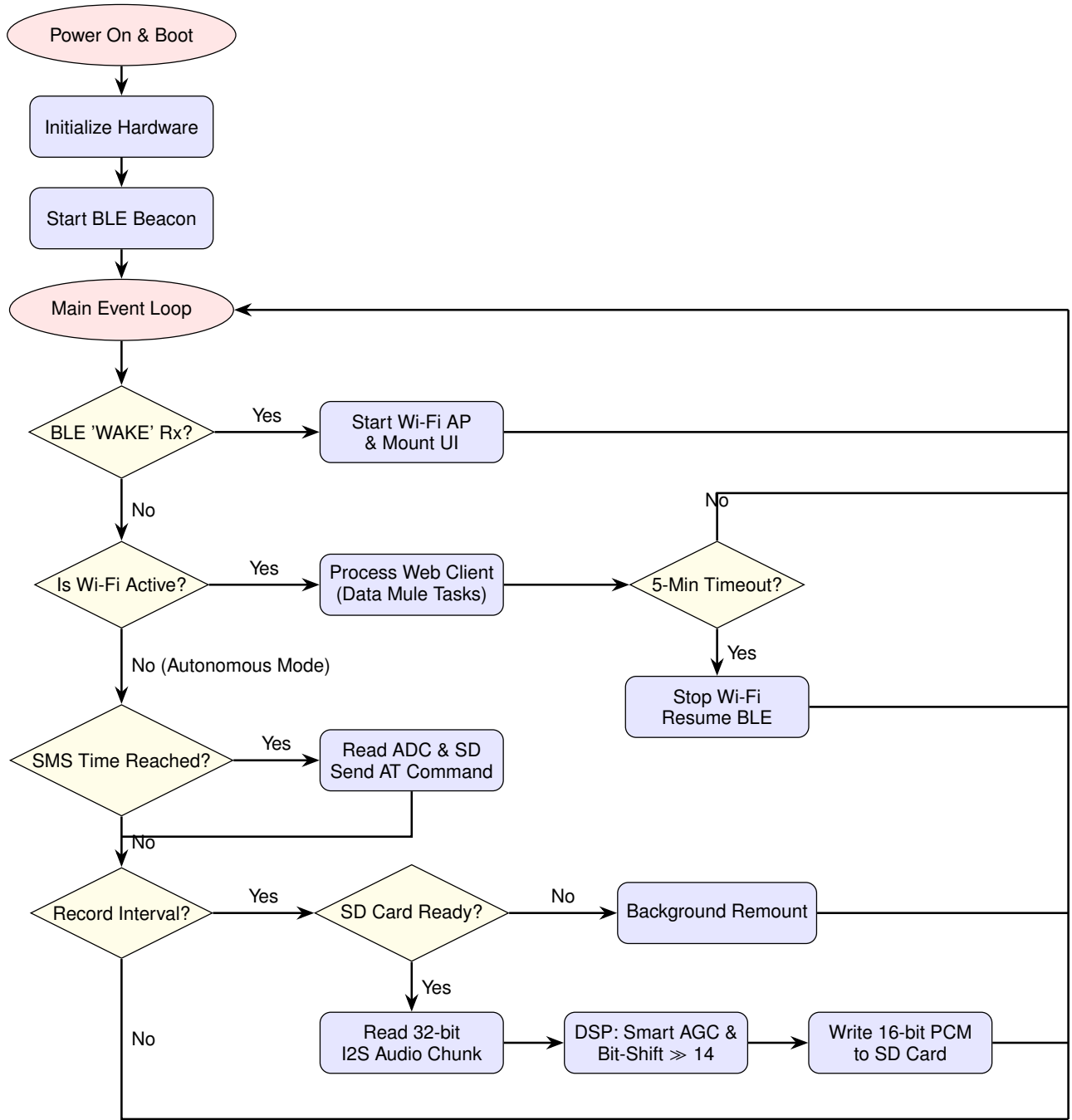


Fig. 3. Algorithmic state flow and logic decision tree for the EcoMule firmware.

physical dynamic range of 87 dB. Consequently, the 16-bit storage container captures 100% of the microphone's usable acoustic data. Discarding the lower bits eliminates invisible thermal noise, halves the SD card storage requirements, and reduces the Wi-Fi transmission time by 50%, significantly extending battery longevity.

VI. CONCLUSION AND FUTURE SCOPE

This project validates a highly resilient paradigm for remote eco-acoustic monitoring. By replacing continuous cloud

streaming with an interactive BLE/Wi-Fi Data Mule interface and SMS telemetry, the energy footprint is vastly minimized. Furthermore, the implementation of localized I2S digital signal processing ensures the acquisition of research-grade acoustic data, effectively bridging the gap between embedded IoT capabilities and the stringent demands of modern ecological conservation.

Scope for Further Improvement:

- **Weatherproofing:** Enclosing the system in an IP67-rated ABS box utilizing PG9 cable glands and ePTFE acoustic



Fig. 4. Conceptual overview of the Data Mule architecture. The Ranger physically bridges the connectivity gap by retrieving bulk environmental data via a localized BLE/Wi-Fi handshake in the deep forest.

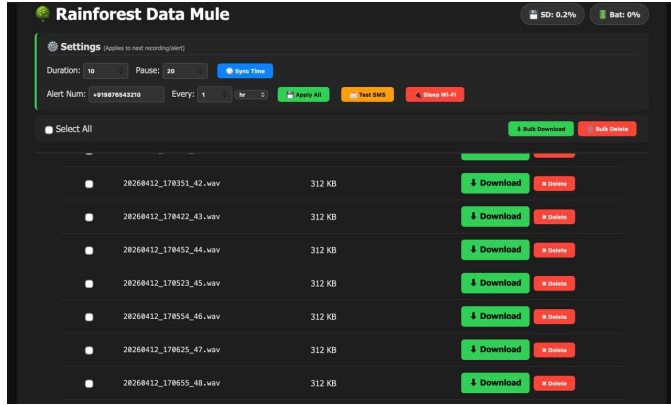


Fig. 5. The elegant, dark-mode Web UI accessed via the local Wi-Fi Access Point, displaying dynamically chunked file lists, live battery/SD metrics, and the Client-Side TAR Bulk Download controls.

membrane patches to fully protect the electronics while capturing pristine audio.

- **On-Device Edge AI:** Integrating lightweight machine learning models (e.g., TensorFlow Lite) for real-time classification of specific events (chainsaws, gunshots) to trigger immediate SMS alerts.

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TABLE I
ESTIMATED SUBSYSTEM POWER CONSUMPTION

Operational State	Current	Voltage	Energy/Day
Acoustic Logging (BLE On)	~ 40 mA	5.0 V	4.80 W h
Data Mule (Wi-Fi Active)	~ 150 mA	5.0 V	0.75 W h
SD Card (Write Bursts)	~ 40 mA	5.0 V	1.20 W h
LTE Modem (SMS Burst)	~ 350 mA	3.8 V	0.05 W h
INMP441 Sensor	~ 2 mA	3.3 V	0.15 W h
Total Estimated Daily Draw			6.95 W h

systems design, which proved pivotal in overcoming the rigorous hardware and software constraints of this project.

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